CS 301: Introduction to Artificial Intelligence

Instructor:	Dr. Sarah Johnson
Email:	s.johnson@university.edu
Office:	Engineering Building, Room 304
Office Hours:	Monday & Wednesday, 2:00 PM - 4:00 PM
Class Time:	Tuesday & Thursday, 10:00 AM - 11:30 AM
Location:	Science Hall, Room 201
Credits:	3 units

Course Description

This course provides a comprehensive introduction to the field of Artificial Intelligence, covering fundamental concepts, algorithms, and applications. Students will explore machine learning, neural networks, natural language processing, and computer vision. Through hands-on projects and programming assignments, students will gain practical experience in implementing AI systems.

Prerequisites

CS 201: Data Structures and Algorithms, MATH 151: Linear Algebra. Proficiency in Python programming is required.

Required Materials

• *Artificial Intelligence: A Modern Approach* by Russell & Norvig (4th Edition)
• Laptop with Python 3.8+ installed
• Access to course website and online learning platform

Learning Objectives

1. Understand fundamental AI concepts including search algorithms, knowledge representation, and reasoning
2. Implement machine learning algorithms including supervised and unsupervised learning techniques
3. Design and train neural networks for various AI applications
4. Apply AI techniques to solve real-world problems in domains such as NLP and computer vision
5. Evaluate ethical implications of AI systems and understand responsible AI development

Grading Policy

Assignment Type	Weight
Homework Assignments (6)	30%
Midterm Exam	25%
Final Project	30%
Class Participation	10%
Quizzes (4)	5%

Grading Scale
A: 93-100, A-: 90-92, B+: 87-89, B: 83-86, B-: 80-82, C+: 77-79, C: 73-76, C-: 70-72, D: 60-69, F: Below 60

Course Schedule

Week	Topics	Assignments
1	Introduction to AI, History and Applications	
2	Search Algorithms: BFS, DFS, A*	HW1 Out
3	Adversarial Search, Game Playing	
4	Knowledge Representation and Logic	HW1 Due, HW2 Out
5	Constraint Satisfaction Problems	
6	Introduction to Machine Learning	HW2 Due, HW3 Out
7	Supervised Learning: Linear/Logistic Regression	
8	Midterm Exam (Week of Oct 15)	HW3 Due
9	Decision Trees and Random Forests	HW4 Out
10	Neural Networks and Backpropagation	
11	Deep Learning: CNNs and RNNs	HW4 Due, HW5 Out
12	Natural Language Processing	Project Proposal Due
13	Computer Vision Applications	HW5 Due, HW6 Out
14	Reinforcement Learning Basics	
15	Ethics in AI, Future Directions	HW6 Due
16	Final Project Presentations	Final Project Due

Important Dates

Event	Date
First Day of Class	September 3, 2024
Last Day to Add/Drop	September 17, 2024
Midterm Exam	October 17, 2024, 10:00 AM - 11:30 AM
Thanksgiving Break	November 27-29, 2024 (No Class)
Final Project Due	December 10, 2024, 11:59 PM
Final Exam Period	December 12-19, 2024

Course Policies

Attendance Policy:
Regular attendance is expected. Students are allowed two unexcused absences. Each additional absence will result in a 2% deduction from the participation grade. If you must miss class, please notify the instructor in advance.

Late Submission Policy:
Homework assignments submitted late will receive a 10% deduction per day, up to 3 days. After 3 days, assignments will not be accepted. Extensions may be granted for documented emergencies with prior approval from the instructor.

Academic Integrity:
All work submitted must be your own. You may discuss concepts with classmates, but all code and written work must be completed individually unless explicitly stated as a group project. Violations of academic integrity will result in a zero on the assignment and possible further disciplinary action.

Accessibility Services:
Students with disabilities who need accommodations should contact the Office of Disability Services within the first two weeks of the semester. All accommodations must be approved through the official process.

Technology in the Classroom:
Laptops and tablets are permitted for note-taking and in-class exercises. Use of social media, email, or other non-class activities during lecture is prohibited and may result in a participation grade reduction.

Communication

Email is the primary method of communication. The instructor will respond to emails within 24 hours on weekdays. For urgent matters, please include "URGENT" in the subject line. Course announcements will be posted on the learning management system.

Additional Resources

- **Office Hours:** Drop-in sessions every Monday and Wednesday, 2:00-4:00 PM
- **Teaching Assistants:** Contact information will be provided on the course website
- **Study Groups:** Self-organized study groups are encouraged
- **Online Forums:** Discussion board available on course website for Q&A;
- **Tutoring Center:** Free tutoring available at the Computer Science Learning Center